

Unit 8, Lesson 6: Evaluating Numeric Expressions – Part 3

Benchmarks

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.NSO.2.2 Add, subtract, multiply and divide rational numbers with procedural fluency.

Learning Target

- ☐ I can evaluate numeric expressions with up to six operations.

8.6.1 Warm-Up: Bell Work

Explain the steps needed to evaluate the expressions.

1. $(10 + 18 \cdot 2)^5 \div 3$

2. $[(7 - 10)^2]^4 \times 2$

8.6.2 Guided Instruction: Evaluating Expressions with Multiple Steps

Two students evaluated the expression below in two different ways. Review the work of each student. Then, answer the questions.

Diego	Valeria
$[0.5 \times (0.5 + 3.5)^2]^3 + 26 \div 13$	$[0.5 \times (0.5 + 3.5)^2]^3 + 26 \div 13$
$[0.5 \times 4^2]^3 + 26 \div 13$	$0.5^3 \times (0.5 + 3.5)^{2 \times 3} + 26 \div 13$
$[0.5 \times 16]^3 + 26 \div 13$	$0.5^3 \times (0.5 + 3.5)^6 + 26 \div 13$
$8^3 + 26 \div 13$	$0.5^3 \times 4^6 + 26 \div 13$
$512 + 26 \div 13$	$0.125 \times 4^6 + 26 \div 13$
$512 + 2$	$0.125 \times 4,096 + 26 \div 13$
514	$512 + 26 \div 13$
	$512 + 2$
	514

1. How were Diego’s steps and Valeria’s steps different from each other?
2. Discuss with your partner whether you think one method was easier to use to evaluate the expression over the other.

8.6.3 Activity: Expression Game

- Each group will use a standard number cube with sides numbered 1 – 6 to play a game. One player rolls the number cube. Each player decides which box to place that number in the expression. Continue rolling the cube until all blanks in the expression are filled. Evaluate your expression and record your score based on the point system in the table. If on any round two players' expressions have the same value, both players **lose 10 points**.

Round 1 Scoring – the player with the greatest value earns 8 points.	$[(_\times__)^2 + __] \div __$	My answer _____ Points earned _____
Round 2 Scoring – the player with the least value earns 5.5 points.	$[(_\times__)^2 + __] \div __$	My answer _____ Points earned _____
Round 3 Scoring – the player with the greatest absolute value loses 3.7 points.	$(_\times__)^2 - (_\div__)$	My answer _____ Points earned _____
Round 4 Scoring – the player with the absolute value closest to zero earns 11 points.	$(_\times__)^2 - (_\div__)$	My answer _____ Points earned _____

My Game Score

Round 1	Round 2	Round 3	Round 4	My Score

2. Discuss with the other players what strategies they used to determine where to put the numbers. Summarize your discussion.

8.6.4 Try It: Evaluating Expressions with Multiple Steps

1. In the table an expression has been evaluated, but the steps are mixed up. Order the expressions using the numbers 1 – 7 on the left. Write an explanation for each step.

Step	Expression	Explain the Step
	$ -11.591 + 15 - (-3.1)^2$	
	$11.591 + 15 - 9.61$	
	$11.591 + 15 - (-3.1)^2$	
1	$ 18.2 + (-3.1)^3 + 15 - (-3.1)^2$	Original expression
	$26.591 - 9.61$	
	$ 18.2 + (-29.791) + 15 - (-3.1)^2$	
	16.981	

2. Evaluate each expression without using a calculator.

a. $\left(\frac{3}{4}\right)^2 \times 32 + 3^2$

b. $[(10 + 4.5)^2 - 10.25]^2 \div 40$

c. $6.7 - (2^3 - 3)^3 \div 10 + 4$

8.6.5 Activity: Class Debate

James wants to choose an expression that has a value of -778 . He can choose between the expressions shown.

Expression A $14 - (10 \times 3^2)^2 + 6 \times 3$	Expression B $14 - 10 \times (3^2)^2 + 6 \times 3$
Expression C $(14 - 10) \times (3^2 + 6)^2 \times 3$	Expression D $14 - [10 \times (3^2)^2] + 6 \times 3$

Which expression should James choose? _____

- Each corner of the room is labeled A, B, C, or D. After selecting the expression that you think James should choose, go to the corner of the room that matches the label for that expression.
- With your classmates who selected the same expression, discuss why your expression is the expression that James should choose. Discuss why James should not choose the other expressions. One member of your group will act as a representative during a class debate.
- For the debate, prepare an opening statement and potential rebuttals for arguments from other groups. Keep arguments and rebuttals centered on the mathematics.
- After the debate, explain whether you would select the same expression again.

8.6.6 Wrap-Up: Ticket Out the Door

1. Evaluate each expression without a calculator. Show each step you use.

$$-\left|5 - \frac{3}{5}\right| + \frac{4}{15}(4^2 - 61)$$

Work

$$\frac{(1.5 + |-3.25 - 4.25|)^3}{12}$$

Work

